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TEAM NAME: **POWERHOUSE**

COLLEGE NAME: **SRI VENKATESWARA COLLEGE OF ENGINEERING**

PROBLEM STATEMENT: **Secure AI-Based Phishing Detection & Prevention System**

PROBLEM STATEMENT

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- Phishing attacks through emails, SMS, and fake websites are rapidly increasing and becoming more sophisticated.

- There is a need for a real-time, adaptive, and intelligent phishing detection system that can accurately identify threats across multiple communication channels.

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SOLUTION

Real-time AI engine analyzes emails, SMS, and websites to instantly detect phishing attempts across all platforms.

Provides instant warnings to users and administrators when phishing activity is detected.

Uses NLP and URL feature extraction to identify suspicious keywords, malicious links, and fake domain patterns.

Continuously re-trains machine learning models to adapt to new and evolving phishing techniques.

Monitors sender behavior and access patterns to detect hidden and zero-day phishing attacks.

Minimizes false alarms using multi-layer analysis and explains why messages are flagged to improve user safety.

FLOW DIAGRAM

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Phishing Detection System Workflow

Data Collection → Preprocessing → AI Detection → Decision & Alert

Data Collection

Collects data from emails, SMS, and URLs

Gathers raw data from various sources

Preprocessing

Cleans and tokenizes text, extracts features

Prepares data for AI analysis

AI Detection

Analyzes content and classifies threats

Uses NLP and ML to identify phishing

Decision & Alert

Generates risk scores and sends alerts

Notifies users and administrators of threats

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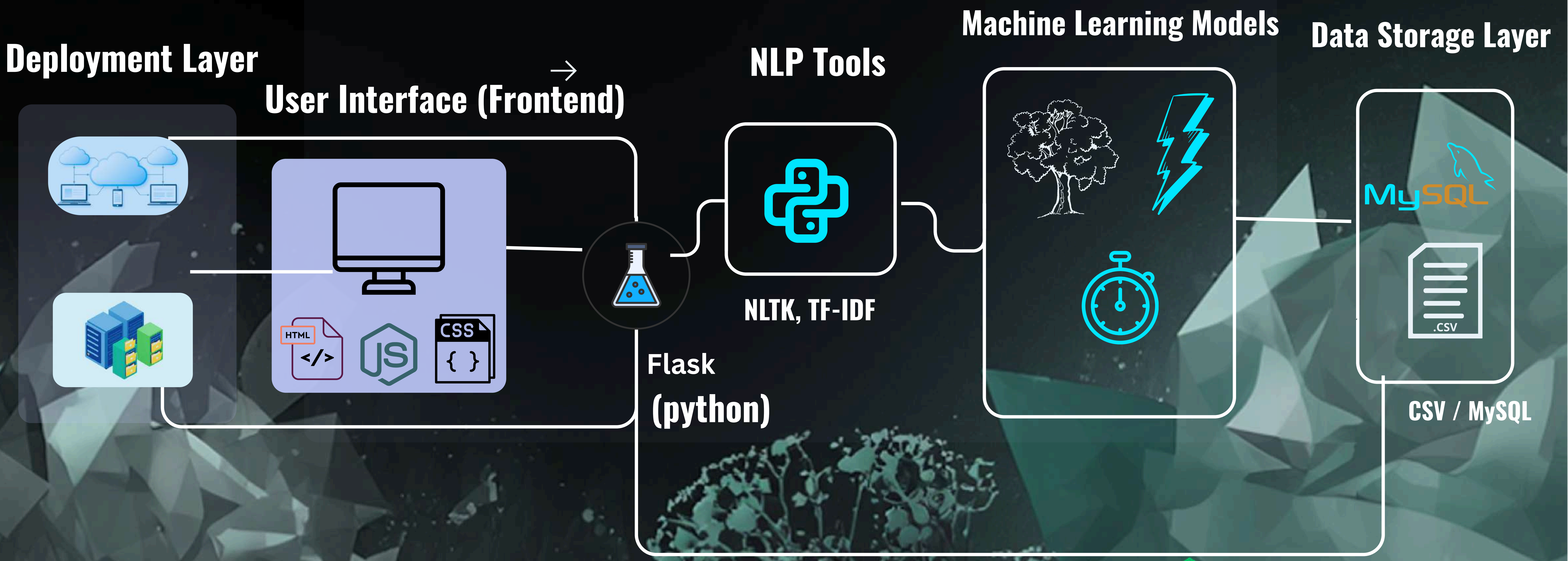
TECHNICAL REQUIREMENTS



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NOVELTY / USE CASE

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- **Adaptive AI-Based Detection:** Uses self-learning machine learning models to identify zero-day and evolving phishing attacks beyond static rules and blacklists.
- **Multi-Channel Unified Protection:** Detects phishing across Email, SMS, and Websites within a single real-time platform.
- **Intelligent Content, URL & Behavior Analysis:** Combines NLP-based message analysis with URL features and sender behavior to improve accuracy and reduce false positives.
- **Real-Time Alerts with User Awareness:** Provides instant alerts along with clear explanations, helping users understand and avoid phishing threats.

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BUSINESS MODEL / TARGET MARKET

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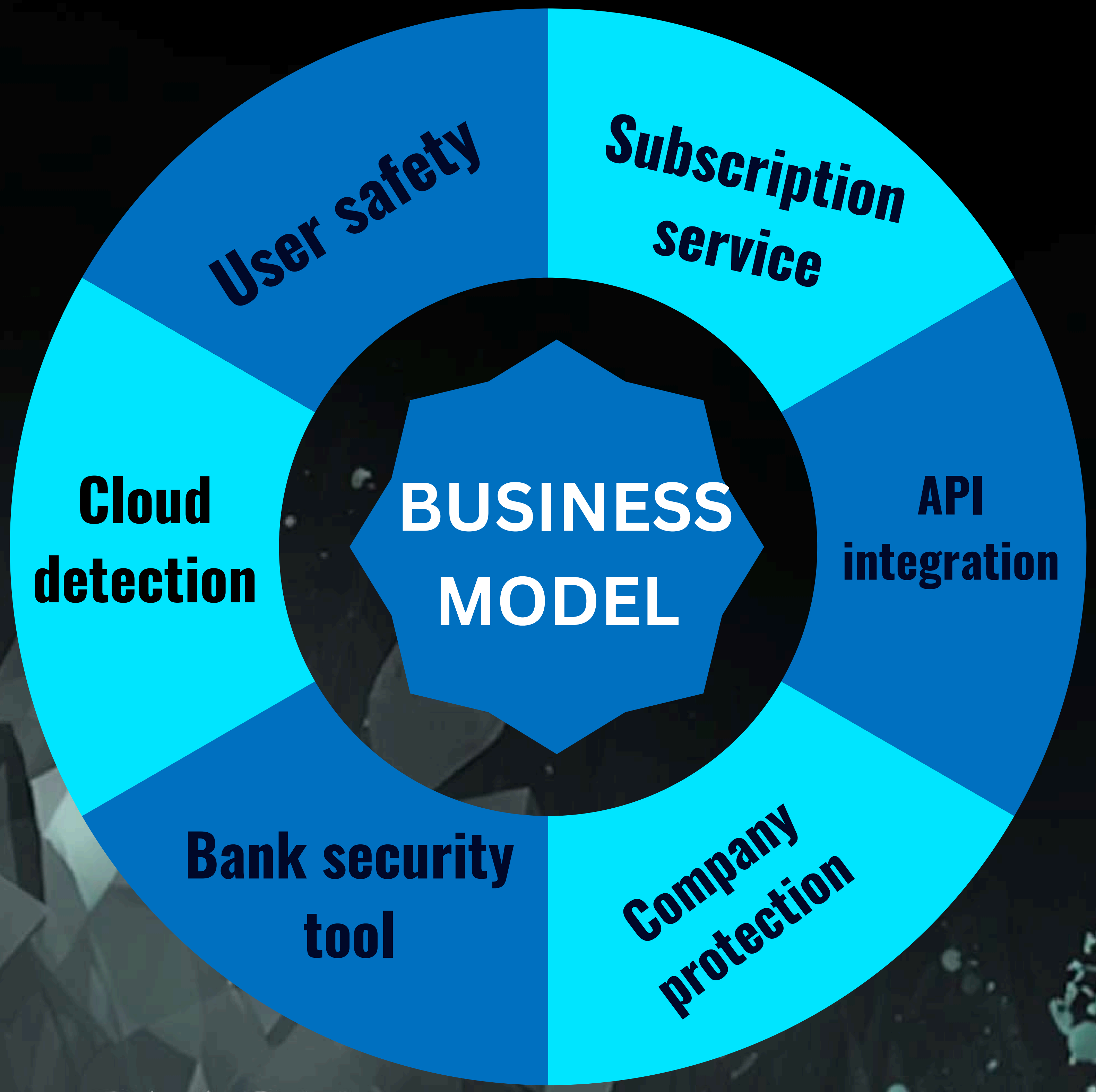

TARGET MARKET

- Banks and financial institutions
- Businesses and corporate organizations
- Regular internet and mobile users

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TEAM DETAILS

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